FORMAL UNRAMIFIEDNESS RESTRICTS TO OPEN SUBSCHEMES OF SOURCE AND BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 855-860 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 862-874, the complete author proof (11 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

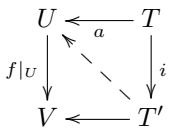
Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The lifting definition of formal unramifiedness (definition-formally-unramified, same chapter); cited input with the uniqueness argument written in the retained span.

Source reference. Original lemma 02H8: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 798–809. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $U \subset X$ and $V \subset S$ be open such that $f(U) \subset V$. If f is formally unramified, so is $f|_U : U \to V$.*

Proof. Consider a solid diagram



as in Definition 02H8. If f is formally ramified, then there exists at most one S-morphism $a' : T' \to X$ such that $a'|_T = a$. Hence clearly there exists at most one such morphism into U. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.